

Inflammation, Senescence and Plaque

IL-6/IL6R, PCR, IL-8/CXCL8, TNF, NLRP3, Lp-PLA2/PLA2G7, HMGB1, GDF-15, MCP-1 and 9p21 (CDKN2B-AS1), CRP scores, IL-6, IL-8, IL-18 and coronary disease, and loci load of inflammatory markers

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Genetic CRP is low to typical (scores in the percentiles 12.0, 25.8 and 25.9; loci load in 40.4). Sérgio F has 1 allele of IL6R (Asp358Ala), which reduces classical inflammatory signaling of IL-6 and is associated with less coronary disease. The **IL-8** score is in percentile 17.4 (low). **TNF –308, NLRP3 Q705K** and **CCL2 –2518** are without risk alleles. Polygenic coronary disease is low (metaGRS in percentile 4.9; GPSmult in 29.7).

Unfavourable. It is homozygous G/G in the locus 9p21 (CDKN2B-AS1/ANRIL, rs10757278), the largest common locus of coronary disease, which acts by proliferation and senescence of arterial wall cells. The loci load of GlycA (acetylated glycoproteins, a low-grade marker of chronic inflammation) is in percentile 90.4 and that of **GDF-15** (senescence marker) in the 83.1. The loci load of IL-8 is in the 98.1, in contradiction with the score (low); CXCL8 –251 is A/A (2 alleles plus IL-8, weak evidence). It has 1 plus **Lp-PLA2** (PLA2G7 R92H) and 1 plus **CRP** in the CRP gene.

What does it mean? The IL-6/CRP axis, the main axis of systemic inflammation, is in favor of Sérgio F, and the total polygenic coronary risk is low despite 9p21 G/G. The unfavourable signs are in the markers of low-grade inflammation and senescence (GlycA and GDF-15) and in the arterial wall (9p21), which respond to exercise, visceral adiposity, sleep and diet. The coronary plaque is also in the Advanced Cardiovascular Report (§26) and immunity in §44.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 12 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 11 of the chip, 1 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 2 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 8 PGS Catalog scores calculated locally with 5.369.289 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 1969 loci, 1678 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all scores of inflammation and coronary disease, a higher percentile = more inflammation or more risk.

Systemic inflammation

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
C-reactive protein	PGS002873 (Ma 2022, PRS-CS, 1.11 million variants)	1.082.460 1.113.831 (weighted coverage 96%)	25,8	TIPIC	typical
C-reactive protein	PGS001946 (Privé 2022, penalized, 71 thousand variants)	68.830 71.444 (weighted coverage 96%)	12,0	BAIXA	favourable
C-reactive protein	PGS000314 (Xie 2020, 77 variants)	76 77 (weighted coverage 99%)	25,9	TIPIC	typical

Cytokines

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Interleucina-6 (IL-6)	PGS000252 (Folkersen 2020, LDpred, 552 thousand variants)	541.930 552.426 (weighted coverage 95%)	46,0	TIPIC	typical
Interleucina-8 (IL-8/ CXCL8)	PGS000254 (Folkersen 2020, LDpred, 552 thousand variants)	541.477 551.990 (weighted coverage 98%)	17,4	BAIXA	favourable
Interleucina-18 (IL-18)	PGS000249 (Folkersen 2020, LDpred, 552 thousand variants)	541.739 552.186 (weighted coverage 98%)	65,4	TIPIC	typical

Coronary plaque

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Coronary disease (metaGRS)	PGS000018 (Inouye 2018, 1.75 million variants)	1.476.469 1.745.179 (weighted coverage 79%)	4,9	MENOS PROVABLE	favourable
Coronary disease (GPSmult)	PGS003725 (Patel 2023, 1.30 million variants)	1.116.308 1.296.172 (weighted coverage 94%)	29,7	TIPIC	typical

4. Key variants by topic

Variants with effect on cytokines, chemokines, Lp-PLA2, allarmins and in the 9p21 locus.

Cytokines and inflammasome

IL-6 is the central cytokine of aging inflammation (inflammation): it stimulates CRP in the liver. The IL6R receptor decides how much of this signaling reaches the cells. IL-8 recruits neutrophils and the inflammasome NLRP3 releases IL-1 β and IL-18.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
IL6R	rs2228145	Asp358Ala	A/C	chip	C (1 copy/s)	favourable	Asp358Ala: C allele (Ala) increases the soluble receptor and reduces classical IL-6 signaling; less CRP and less coronary disease (IL6R MR Consortium 2012)	Strong
IL6	rs1800795	–174 G>C	G/G	chip	C (0 copy/s)	without judgment	–174 G>C promoter; inconsistent effect on IL-6 among studies	Weak
CRP	rs2794520		C/T	chip	C (1 copy/s)	unfavourable (light)	CRP gene regulator: the effect allele increases basal CRP (genetic CRP is not causal to coronary disease) [GWAS Catalog: C-reactive protein levels, allele C, beta 0.182 [0.17-0.19] unit increase, p=4E-523, PMID 30388399]	Strong
CXCL8 (IL-8)	rs4073	–251 T>A	A/A	chip	A (2 copy/s)	unfavourable	–251 T>A promoter: allele A is associated with more production of IL-8 by neutrophils (small association studies)	Weak
ACKR1 (Duffy)	rs12075	Asp42Gly	A/A	chip	G (0 copy/s)	without judgment	Asp42Gly (FYA/FYB): the Duffy receptor "sponge" of chemokines; changes circulating levels of IL-8 and MCP-1 without clear effect on tissue inflammation	Moderate
TNF	rs1800629	–308 G>A	G/G	chip	A (0 copy/s)	favourable	–308 G>A promoter: allele A is associated with more TNF- α in some studies	Weak
NLRP3	rs35829419	Q705K	C/C	chip	A (0 copy/s)	favourable	Gln705Lys: NLRP3 inflammasome function gain (plus IL-1 β and IL-18)	Weak

Plate: Lp-PLA2, chemokines and 9p21

Lp-PLA2 (PLA2G7 gene) travels in LDL and generates oxidized phospholipids in the plate. MCP-1 recruits monocytes to the arterial wall. The locus 9p21 is the largest common locus of coronary disease and acts on the vessel wall (proliferation and senescence), not through LDL.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PLA2G7 (Lp-PLA2)	rs1805017	R92H	C/T	chip	T (1 copy/s)	unfavourable (light)	Arg92His: T allele increases the activity and mass of Lp-PLA2 (the enzyme that oxidizes phospholipids in the LDL plate) [GWAS Catalog: Lipoprotein-associated phospholipase A2 activity and mass, allele T, beta 0.043 [0.035-0.051] unit increase, p=2E-23, PMID 22003152]	Moderate
CCL2 (MCP-1)	rs1024611	–2518 A>G	A/A	chip	G (0 copy/s)	favourable	–2518 A>G promoter: G allele increases MCP-1, which recruits monocytes to the arterial wall	Weak

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CDKN2B-AS1 (9p21, ANRIL)	rs10757278	9p21	G/G	chip	G (2 copy/s)	unfavourable	Locus 9p21: the largest common locus of coronary disease (OR ~1.3 per allele); regulates p16/p15 (CDKN2A/B), senescence and proliferation of smooth muscle cells	Strong

Senescence and alarmins (HMGB1, GDF-15)

HMGB1 is an allarmine (DAMP) released by damaged or senescent cells. GDF-15 is a marker of the secretory phenotype of senescence and mitochondrial stress; it rises with age.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NLRP12 (HMGB1)	rs62143206		G/G	chip	T (0 copy/s)	favourable	Locus with the highest effect on HMGB1 levels measured by aptamer (can be binding artifact) [GWAS Catalog: High mobility group protein B1 levels (HMGB1.2524.56.3), T allele, beta 0.680 [0.62-0.74] unit increase, p=2E-132, PMID 29875488]	Weak
GDF15	rs1227734		C/T	imputation DR2=0.81	T (1 copy/s)	without judgment	Locus GDF15: levels of GDF-15, a marker of the secretory phenotype of senescence and mitochondrial stress (genetic GDF-15 is not causal to the disease) [GWAS Catalog: Growth difference factor-15 levels, T allele, beta 0.374 unit increase, p=3E-250, PMID 33067605]	Moderate

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each inflammatory marker (CRP, IL-6, IL-8, Lp-PLA2, GlycA, GDF-15, MCP-1 and HMGB1), with the alleles oriented towards the highest value, compared to the reference population. Many of these loci are pQTL measured by immunoassay or aptamer and may reflect the binding to the reagent and not the real concentration.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
C-reactive protein	1481 (1241)	1238 de 2482	1238	40,4	typical
IL-6	15 (13)	12 de 26	13	47,6	typical
IL-8 (CXCL8)	18 (16)	20 de 32	16	98,1	unfavourable
Lp-PLA2 (PLA2G7)	48 (40)	39 de 80	39	59,1	typical
GlycA (acetylated glycoproteins)	342 (313)	335 de 626	313	90,4	unfavourable
GDF-15 (senescence)	28 (19)	20 de 38	19	83,1	unfavourable
MCP-1 (CCL2)	35 (34)	26 de 68	34	49,1	typical
HMGB1	2 (2)	1 de 4	2	47,0	typical

6. Favourable

- **Low to typical CRP:** scores in the percentiles 12.0, 25.8 and 25.9; loci load in the 40.4. **Moderate.**
- **IL6R Asp358Ala A/C** (1 allele Ala): less classical IL-6 signaling, less CRP and less coronary disease (the same effect of tocilizumab, in small). **Strong.**
- **Low polygenic coronary disease:** metaGRS in percentile 4.9 (79% coverage), GPSmult in the 29.7. **Moderate.**
- **IL-8** with a score in percentile 17.4; Typical **IL-6** (46.0 score, 47.6 loci). **Moderate.**
- **TNF –308 G/G, NLRP3 Q705K C/C, CCL2 –2518 A/A:** no alleles of more TNF, more inflammasome or more MCP-1; MCP-1 (loci) in percentile 49.1. **Weak.**
- **HMGB1:** without the allele of more HMGB1 of NLRP12; loci in percentile 47.0. **Weak.**

7. Unfavourable

- **9p21 rs10757278 G/G**: 2 risk alleles in the largest common locus of coronary disease (OR ~1.3 per allele); acts by senescence and proliferation of smooth muscle cells (CDKN2A/B). It is compensated in the total score, which is low. Strong for the locus.
- **GlycA** (loci) in percentile 90.4: genetic trend towards more low-grade chronic inflammation. **Moderate.**
- **GDF-15** (loci) in percentile 83.1: plus basal GDF-15 (senescence marker and mitochondrial stress; genetic GDF-15 is not causal). **Weak moderated.**
- **IL-8**: loci load at percentile 98.1 and CXCL8 –251 A/A, against the score no 17.4; mixed signal, probably affected by the binding loci (Duffy). **Weak.**
- **PLA2G7 R92H C/T** (1 allele of more activity of Lp-PLA2; loci in 59.1) and **CRP rs2794520 C/T** (1 allele of more CRP). **Moderate.**
- **IL-18** with a score in percentile 65.4 (typical-high). **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Inflammaging	IL-6/CRP axis favourable (IL6R Ala, CRP low genetic); GlycA (loci) in percentile 90.4	Mixed
Chronic Inflammation	High GlycA in loci; TNF, NLRP3 and CCL2 without risk alleles	Mixed
Cell Senescence	9p21 G/G (CDKN2A/B, p16) and GDF-15 (loci) in percentile 83.1	Unfavourable
Vascular Aging	Low polygenic coronary disease (percentile 4.9), despite 9p21 G/G	Favourable
Intercellular Communication	IL6R Asp358Ala: less classical IL-6 signaling	Favourable
Release of Mitochondrial DAMPs	HMGB1 typical; GDF-15 (mitochondrial stress) with high trend	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Fiber at maximum** (legumes, oats, seeds, vegetables), extra virgin olive oil, red fruits and spices (curcuma, ginger): lower GlycA and CRP. *Suggestion (not applied)*.
2. **Omega-3 3.2-4 g/day (already in the plane)**: low triglycerides and generates pro-resolution mediators (resolvinas); maintain. *Suggestion (not applied)*.
3. **Regular exercise (aerobic + force)**: the factor that lowers GlycA, chronic IL-6 and the load of senescent cells; interrupts sitting time. *Suggestion (not applied)*.
4. **Low visceral fat** (treatment with ongoing retatrutide helps; see the Inflammatory Fat and Lipolysis report): visceral adipose tissue is the largest source of IL-6 and GlycA. *Suggestion (not applied)*.
5. **9p21 G/G and board**: LDL and ApoB as low as possible; consider with the physician a coronary calcium score as the base image of the plaque, since 9p21 acts on the arterial wall and not on LDL. *Suggestion (not applied)*.
6. **Measure once the high sensitivity CRP** (and, if available, GlycA) out of infections, to confirm what the genes predict. *Suggestion (not applied)*.
7. **Sources of chronic inflammation**: oral health (periodontitis up to IL-6 and CRP), sleep of 7-9 h, zero tobacco and zero alcohol (already fulfills); regular breakfast as an option (low to CRP in observational studies). *Suggestion (not applied)*.
8. **Senolytics** (dasatinib + quercetin, physetin): continue to be experimental; they are not suggested outside the clinical trial. *Suggestion (not applied)*.

10. Limitations

- The loci of IL-8, MCP-1, GDF-15 and HMGB1 come in most studies of proteomics (immunoassay and aptamer); some reflect the binding to the reagent (e.g. Duffy/ACKR1) and not the real concentration.
- Genetic levels of CRP and GDF-15 are not causal for coronary disease (mendelian randomization studies); IL6R and 9p21 are.
- The metaGRS score has 79% of weighted coverage.
- There is no robust common variant in the HMGB1 gene itself; genetic HMGB1 is little informative.
- The load of SNPs in linkage disequilibrium is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palindromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (8 scores: CRP, IL-6, IL-8, IL-18 and coronary disease). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510-8$), alleles guided by the published effect signal, empirical calibration in the same reference population.
- **Interpretation:** reference literature (IL6R Genetics Consortium 2012; Helgadottir 2007 and McPherson 2007 for 9p21; Ridker 2017 for the IL-1/IL-6 pathway; Folkersen 2020 for inflammatory proteins), GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Cytokines and inflammasome	IL6R	rs2228145	A/C	chip	C	1	strand +	1 copy of the protective allele
Cytokines and inflammasome	IL6	rs1800795	G/G	chip	C	0	strand + (palindromic; assumed +)	0 copy of the effect allele
Cytokines and inflammasome	CRP	rs2794520	C/T	chip	C	1	strand +	1 copy of risk allele
Cytokines and inflammasome	CXCL8 (IL-8)	rs4073	A/A	chip	A	2	strand + (palindromic; assumed +)	2 risk allele copies
Cytokines and inflammasome	ACKR1 (Duffy)	rs12075	A/A	chip	G	0	strand +	0 copy of the effect allele
Cytokines and inflammasome	TNF	rs1800629	G/G	chip	A	0	strand +	without the risk allele
Cytokines and inflammasome	NLRP3	rs35829419	C/C	chip	A	0	strand +	without the risk allele
Plate: Lp-PLA2, chemokines and 9p21	PLA2G7 (Lp-PLA2)	rs1805017	C/T	chip	T	1	strand +	1 copy of risk allele
Plate: Lp-PLA2, chemokines and 9p21	CCL2 (MCP-1)	rs1024611	A/A	chip	G	0	strand +	without the risk allele
Plate: Lp-PLA2, chemokines and 9p21	CDKN2B-AS1 (9p21, ANRIL)	rs10757278	G/G	chip	G	2	strand +	2 risk allele copies
Senescence and alarmins (HMGB1, GDF-15)	NLRP12 (HMGB1)	rs62143206	G/G	chip	T	0	strand +	without the risk allele
Senescence and alarmins (HMGB1, GDF-15)	GDF15	rs1227734	C/T	imputation DR2=0.81	T	1	strand +	1 copy of effect allele